

Exercise series 4: Rel. Algebra To SQL

Introduction to databases



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From Rel. Algebra to SQL Queries

Everything you can do using relational algebra can be done using SQL

Primitive operations in Rel. Alg:

- **projection** (π)
- **selection** (σ)
- **rename** (ρ)
- **cartesian product** ($\times$)
- **union** ($\cup$)
- **subtract** ($-$)

Composite operations in Rel. Alg:

- **intersect** ($\cap$)
- **join** ($\bowtie$)
- **division** ($\div$)



PC(model, price) Laptop(model, price)	Makes(name, model) Producer(name)
Projection (π)	
π price (Laptop)	SELECT price FROM Laptop ;
Selection (σ)	
σ price > 2000 (PC)	SELECT * FROM PC WHERE price > 2000 ;
Rename (ρ)	
ρ model2 <- model (Laptop)	SELECT model AS model2, price FROM PC;
Cartesian product ($\times$)	
Laptop $\times$ PC	SELECT * FROM Laptop, PC ;
Union ($\cup$)	
Laptop $\cup$ PC	SELECT * FROM Laptop UNION SELECT * FROM PC ;
Substract (-)	
Laptop - PC	SELECT * FROM Laptop EXCEPT SELECT * FROM PC ;

PC(model, price) Laptop(model, price)	Makes(name, model) Producer(name)
Intersect ($\cap$)	
PC $\cap$ Laptop	SELECT * FROM PC INTERSECT SELECT * FROM Laptop ;
As composition Laptop – (Laptop – PC)	
Join ($\bowtie$)	
PC $\bowtie$ Producer	SELECT * FROM PC NATURAL JOIN Makes ;
As composition σ PC.model=Makes.model (PC $\times$ Makes)	
Division ($\div$)	
Makes $\div$ Producer	-- MUST USE COMPOSITION FROM PRIMITIVE OPERATIONS
As composition π model (Makes) – π model ((π model (Makes) $\times$ Producer) – Makes)	



Problem 1: PCs, Laptops and Printers

Consider the example data provided here: <https://www.db-fiddle.com/f/mazcNfJRGk7SyQ2yru44sj/1>
(yes, it is the same of the past session about relational algebra)

- Laptop (model: int, speed: float, ram: int, hd: int, screen: float, price: int)
- PC (model: int, speed: float, ram: int, hd: int, price: int)
- Printer (model: int, color: boolean, type: string, price: int)
- Product (maker: char, model: int, type: string)



Problem 1: PCs, Laptops and Printers

PC.model	PC.speed	PC.ram	PC.hd	PC.price
1001	2.66	1024	250	2114
1002	2.1	512	250	995
1003	1.42	512	80	478
1004	2.8	1024	250	649
1005	3.2	512	250	630
1006	3.2	1024	320	1049
1007	2.2	1024	200	510
1008	2.2	2048	250	770
1009	2	1024	250	650
1010	2.8	2048	300	770
1011	1.86	2048	160	959
1012	2.8	1024	160	649
1013	3.06	512	80	529

Printer.model	Printer.color	Printer.type	Printer.price
3001	TRUE	'ink-jet'	99
3002	FALSE	'laser'	239
3003	TRUE	'laser'	899
3004	TRUE	'ink-jet'	120
3005	FALSE	'laser'	120
3006	TRUE	'ink-jet'	100
3007	TRUE	'laser'	200

Laptop.model	Laptop.speed	Laptop.ram	Laptop.hd	Laptop.screen	Laptop.price
2001	2	2048	240	20.1	3673
2002	1.73	1024	80	17	949
2003	1.8	512	60	15.4	549
2004	2	512	60	13.3	1150
2005	2.16	1024	120	17	2500
2006	2	2048	80	15.4	1700
2007	1.83	1024	120	13.3	1429
2008	1.6	1024	100	15.4	900
2009	1.6	512	80	14.1	680
2010	2	2048	160	15.4	2300

Product.maker	Product.model	Product.type
'A'	1001	'pc'
'A'	1002	'pc'
'A'	1003	'pc'
'A'	2004	'laptop'
'A'	2005	'laptop'
'A'	2006	'laptop'
'B'	1004	'pc'
'B'	1005	'pc'
'B'	1006	'pc'
'B'	2007	'laptop'
'C'	1007	'pc'
'D'	1008	'pc'
'D'	1009	'pc'
'D'	1010	'pc'
'D'	3004	'printer'
'D'	3005	'printer'
'E'	1011	'pc'
'E'	1012	'pc'
'E'	1013	'pc'
'E'	2001	'laptop'
'E'	2002	'laptop'
'E'	2003	'laptop'
'E'	3001	'printer'
'E'	3002	'printer'
'E'	3003	'printer'
'F'	2008	'laptop'
'F'	2009	'laptop'
'G'	2010	'laptop'
'H'	3006	'printer'
'H'	3007	'printer'



Problem 1: PCs, Laptops and Printers

- Find all makers that do not make laptops or printers.
- Find the manufacturers who sell two or more models of PC.
- Find all makers that sell PC's or Laptop's with at least 2 Gb of ram.
- Find the model with the highest price.



- Find all makers that do not make laptops or printers.

Rel. Algebra	Exact Translation	Preferable Answer
π maker Product - π maker σ (type = 'laptop' $\vee$ type = 'printer') Product	<pre> SELECT maker FROM Product EXCEPT SELECT maker FROM (SELECT * FROM Product WHERE type = 'laptop' OR type = 'printer') AS Subquery; </pre>	<pre> SELECT DISTINCT maker FROM Product P WHERE NOT EXISTS (SELECT * FROM Product S WHERE S.maker = P.maker AND (S.type = 'laptop' OR S.type = 'printer')); </pre>

- Find the manufacturers who sell two or more models of PC.

Rel. Algebra	Exact Translation	Preferable Answer
$\text{Maker_model} =$ $\pi \text{ maker, model}$ $($ $\quad \sigma \text{ type} = \text{'pc'} \text{ Product}$ $)$ $\pi \text{ maker}$ $($ $\quad \sigma \text{ model2} < \text{model}$ $\quad ($ $\quad \quad \text{Maker_model}$ $\quad \quad \bowtie$ $\quad \quad \rho \text{ model2} \leftarrow \text{model}$ $\quad \quad \text{Maker_model}$ $\quad)$ $)$	<pre> CREATE VIEW Maker_and_model AS SELECT maker, model FROM Product WHERE type = 'pc'; SELECT DISTINCT maker FROM Maker_and_model NATURAL JOIN (SELECT maker, model AS model2 FROM Maker_and_model) AS Subquery WHERE model2 < model; </pre>	<pre> CREATE VIEW Maker_and_model AS SELECT maker, model FROM Product WHERE type = 'pc'; SELECT DISTINCT M1.maker FROM Maker_and_model M1, Maker_and_model M2 WHERE M1.maker = M2.maker AND M1.model < M2.model; </pre>

- Find all makers that sell PC's or Laptop's with at least 2 Gb of ram.

Rel. Algebra	Exact Translation	Preferable Answer
π maker (Product $\bowtie$ σ ram $\geq$ 2000 (π model, ram PC $\cup$ π model, ram Laptop))	SELECT DISTINCT maker FROM Product P NATURAL JOIN (SELECT * FROM (SELECT model, ram FROM PC UNION SELECT model, ram FROM Laptop) AS Subsubquery WHERE ram $\geq$ 2000) AS Subquery;	SELECT maker FROM Product P WHERE model IN (SELECT model FROM PC WHERE ram $\geq$ 2000 UNION SELECT model FROM Laptop WHERE ram $\geq$ 2000);

- Find the model with the highest price.

Rel. Algebra	Preferable Answer	
<pre> Model_with_price = π model, price PC $\cup$ π model, price Laptop $\cup$ π model, price Printer All_prices = π price Model_with_price π model (Model_with_price $\bowtie$ (All_prices - π price σ price < price2 (All_prices $\times$ ρ price2 $\leftarrow$ price All_prices))) </pre>	<pre> CREATE VIEW Model_with_price AS (SELECT model, price FROM PC UNION SELECT model, price FROM Laptop UNION SELECT model, price FROM Printer); CREATE VIEW All_Prices AS SELECT price FROM Model_with_price; </pre>	<pre> SELECT model FROM Model_with_price WHERE price IN (SELECT * FROM All_prices EXCEPT SELECT P1.price FROM All_prices P1, All_prices P2 WHERE P1.price < P2.price); </pre>